Safety stop valve 543-010
Instruction manual

Please read the instruction manual completely and retain it for future reference!

Imprint

Safety stop valve 543-010

Instruction manual

Version 1.0

Issued by:

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Installationdata

NOTE Please fill in this form following commissioning. It will help you and your Grundfos Alldos service partner adjust the device during subsequent corrections.

Owner:	
Grundfos Alldos customer no.:	
Contract no.:	
Order no. of device:	
Serial no. of device:	
Commissioned on:	
Location of device:	
Used for:	

Installation diagram

Table of Contents

1 General

1.1 About this manual

In this manual you can find all important information for operating the device described:

- ❑ Technical data
- ❑ Instructions for commissioning and maintenance
- ❑ Safety instructions

If you require further information or if any problems arise that are not described in sufficient detail in this document, please contact GRUNDFOS ALLDOS directly.

1.2 Using this document

- ❑ Descriptions are written in standard continuous text.
- ❑ Lists are identified by squares (❑), subdivisions by dashes (-).
- ❑ Steps requiring user action are identified by bold dots (●), subdivisions by smaller dots (•).
- ❑ Cross-references are indicated by *italic letters* and an arrow (➜).
- ❑ The safety instructions ***WARNING***, ***CAUTION*** and ***NOTE*** have the following meanings:

WARNING ***Risk of injuries and accidents!***

CAUTION ***Risk of malfunctioning or damage to the device!***

NOTE This is a special case.

1.3 Guarantee

A guarantee claim in the sense of our general conditions of sale and delivery can only be recognised if:

- ❑ the device has been used according to the information in this instruction manual,
- ❑ the device has not been opened or incorrectly handled in any manner,
- ❑ any repairs have been carried out solely by authorized specialists,
- ❑ only original parts have been used during repairs,
- ❑ only components approved by GRUNDFOS ALLDOS are used throughout the entire system.

Typical parts subject to wear are excluded from the guarantee, e.g.

- ❑ O-rings and pressure springs

1.4 Explanation of terms

The device in question is described in DIN 19606 as a pressure relief valve. In this manual we refer to it as a „safety stop valve“, as this describes its actual functionality.

2 Safety information

2.1 Use of the device

The safety stop valve 543 prevents pressure from building up in the vacuum line to the dosing regulator and injector within the scope of the potential usage described in this instruction manual.

WARNING	***Other applications are considered as non-approved, and are not permissible. GRUNDFOS ALLDOS cannot be held liable for any damage resulting from such use.***

2.2 Obligations on the part of the operator

The operator of the system is responsible for

- ❑ compliance with country-specific safety regulations
- ❑ training of operating personnel
- ❑ provision of prescribed protective equipment
- ❑ implementation of regular maintenance.

2.3 Avoidance of Danger

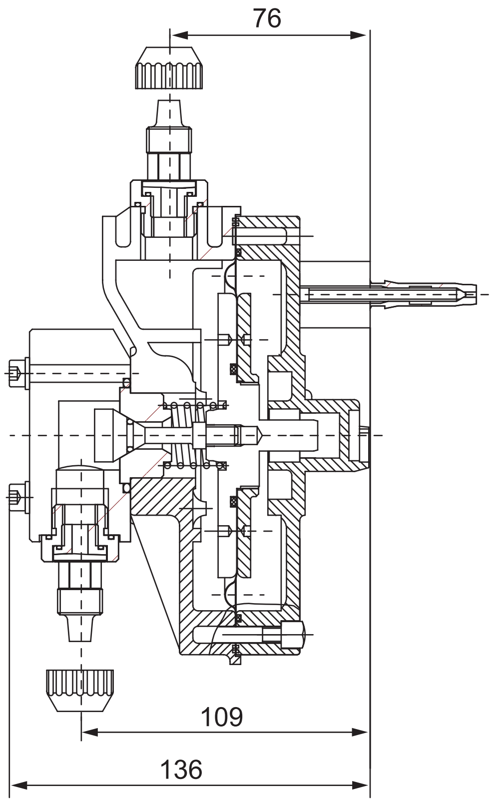
WARNING	***Do not open the device!*** ***Cleaning, maintenance and repairs should only ever be performed by authorised specialist staff!*** ***Observe the section on „Operation and safety of chlorine systems“!***

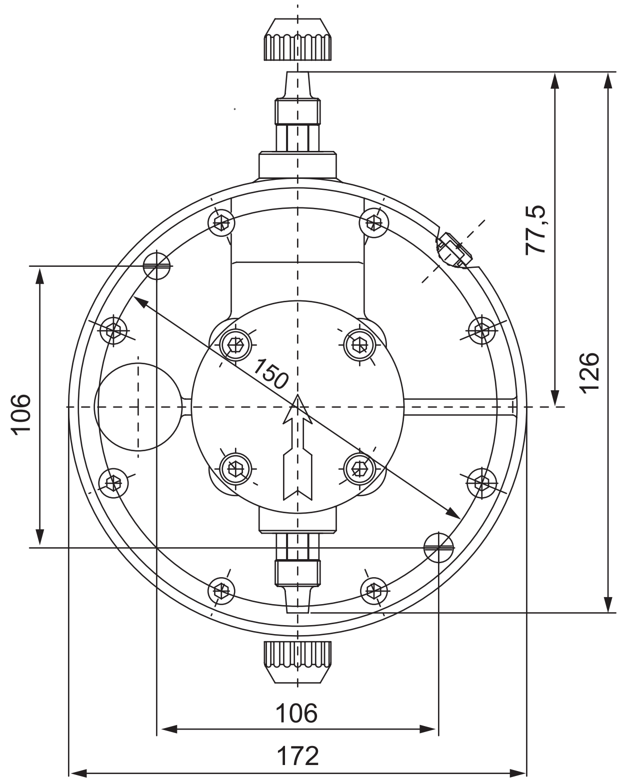
3 Technical data

Dosing medium Chlorine; in special versions also sulphur dioxide and ammonia

Maximum throughput	10 kg/h Cl_2
	10 kg/h SO_2
	5 kg/h NH_3
Connections	For PE tube 10x14
	For PE tube 12x16
	For PVC pipe DN10
	For PVC pipe DN15
	For PE hose 1/2"
	For PE tube 8x11
Wall mounting	direct or with base plate
Materials	PVC, FPM, PVDF, FEP and Hastelloy
Weight	1kg
Pressure loss	about 0.02 bar
Storage temperature	0 to + 50°C
Operating temperature	+ 5 to + 40°C

3.1 Dimensional drawings





4 Installation

4.1 Transport and storage

- Transport the safety stop valve carefully. Never throw it!
- ❑ Dry, cool storage location.

4.2 Unpacking

Note when unpacking

- Make sure you do not discard any accesssories by mistake when disposing of packaging materials.
- Do not let any moisture penetrate the safety stop valve.
- Do not let any foreign bodies penetrate parts that carry gas.
- Assemble as soon as possible after unpacking.

4.3 Assembly

- The safety stop valve is attached to the wall using the screws and plugs supplied, either directly or using the base plate (optional).
- We recommend fitting it in the vicinity of the hole in the wall that passes through to the next room for the vacuum line connected to the dosing regulator and diaphragm non-return on the injector.
- The vacuum dosing line coming from the vacuum regulators or the vacuum change-over device should be connected to the input connection (on the small housing diameter) of the safety stop valve.

NOTE ***The flow direction of the gas is marked by a stick-on red arrow.***

- The continued dosing line that runs to the dosing regulator and diaphragm non-return should be connected to the output connection (on the large housing diameter) of the safety stop valve.

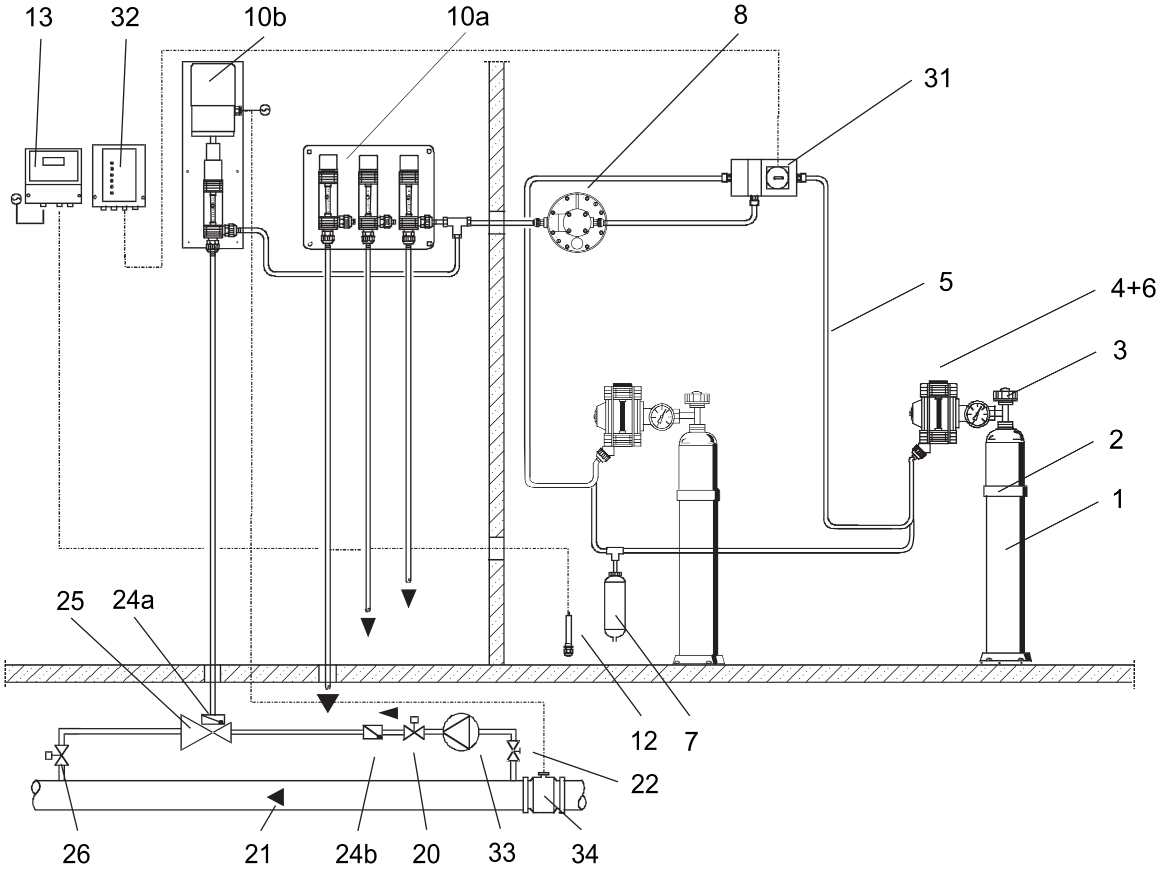
WARNING ***The plastic screw connections should be tightened without using tools.***

- The diaphragm of the safety stop valve is generously proportioned. This means that the pressure loss is minimal and the stop valve acts as a length of pipe when the system is in operation.

NOTE ***The vacuum dosing line should be kept as short as possible to keep pressure losses to a minimum.***

WARNING ***When laying the vacuum dosing line (made of PE hose), any curves should be kept as wide-radius as possible. This prevents the PE hose from getting kinked or damaged in the long term.***

4.4 Installation diagram



1 Chlorine gas container

2 Chlorine gas container retaining clamp

3 Chlorine gas container isolating valve

4+6 Vacuum regulator (pressure relief valve integrated)

5 Vacuum line

7 Adsorption device

8 Safety stop valve

10a Dosing regulator (multiple)

10b Dosing regulator with el. servomotor

12 Sensor for chlorine gas detector

13 Chlorine gas detector

20 Solenoid valve

21 Operating water

22 Isolating valve

24a, 24b Check valve

25 Injector

26 Injection unit

31 Vacuum change-over device

32 Display device for empty message

33 Booster pump

34 Flow meter with current signal output

5 Commissioning

Once all installation work has been completed and following a comprehensive tightness inspection, the safety stop valve can be put into operation.

WARNING ***Commissioning should be peformed solely by specialist authorised staff, who must strictly observe the regulations governing handling of the dangerous chemical chlorine.***

WARNING ***Before commissioning, a tightness test must be performed on all parts that carry chlorine gas. While it is being performed, all staff involved must wear protective clothing and gas masks!***

5.1 Checking the vacuum side for tightness

- Close bottle valves or barrel/bottle connection valves.
- Start up the injector. The dosing lines will be evacuated.
- After around 10 minutes (depending on the length of the dosing lines and the setting of the adjustment spindle on the dosing regulator), no further flow should be indicated at the dosing regulator meter tube.
- However, if the meter tube still indicates a flow, the entire vacuum system must be checked for leaks.

WARNING ***To perform a tightness test, the entire chlorine gas system must be evacuated and free of chlorine gas. The injector must be in operation.***

- To check the individual system components, disconnect the vacuum dosing lines between the vacuum and dosing regulators one after the other and form a seal with your finger.

  If the meter tube indicates flow has stopped after disconnecting a dosing line, the leak is within the last section disconnected.

5.2 Checking the pressure side for tightness

- Stop the injector.
- Open the bottle/barrel valves and/or barrel/bottle connection valves by around 1–2 revolutions until you can hear chlorine gas flowing in.
- Check the valves, screw connections and inlet valves of the vacuum regulator for tightness using ammonia.
- Localise any leaks by guiding the open bottle of ammonia past the relevant sections and pressing out ammonia gas by gently squeezing the plastic bottle.
  Any sections with chlorine gas leaks will form a white mist by reacting with the ammonia gas.

WARNING ***Even the smallest of leaks detected must be dealt with immediately, as corrosion can otherwise occur extremely quickly!***

5.3 Commissioning (initial startup)

Once the tightness tests have been successfully completed, commissioning can be performed:

- Close the adjustment spindle on the dosing regulator.
- Open the isolating valve on the injection unit.
- Open the industrial water valve.
- Switch on the injector driving water.
- Open the valves and/or connection valves of the gas tanks.
- Slowly open the adjusting spindle on the dosing regulator until the suspended part in the meter tube displays the desired stream of chlorine gas.

NOTE **The safety stop valve works automatically. No operator input is required.**

5.4 Taking out of service

5.4.1 Taking out of service temporarily

If the safety stop valve is to be taken out of operation for a short time, it can simply be left in place between the vacuum dosing lines.

If the vacuum dosing lines are removed, the input and output connections of the safety stop valve must be hermetically sealed immediately using suitable means to ensure that no moisture can get in.

WARNING ***Residual chlorine gas and moisture in the air cause the metal components of the safety stop valve to rust.***

5.4.2 Taking out of service long-term

- If the safety stop valve is not going to be used for a long period of time, it should be unscrewed from the wall, disassembled and its components cleaned in warm water.
- Once completely dry, the stop valve should be refitted and the input and output connections must be must be hermetically sealed immediately. No moisture or foreign bodies should be allowed to penetrate the valve during storage, right up to the point when it is fitted back in the vacuum dosing lines.

6 Operation

6.1 Basics

The latest version of DIN 19606, which is dated June 2006 and deals with chlorine gas systems for water treatment, prescribes a pressure relief valve in all cases when the injector is outside the chlorine gas storage room.

NOTE ***The „chlorine gas storage room“ (previously referred to as chlorine gas room) is classed as the room in which the ready-to-use chlorine tanks connected to the chlorine system are located.***

WARNING ***In the event of a malfunction, chlorine gas can escape in this „chlorine gas storage room“. The room must therefore be monitored with a gas detector.***

Vacuum chlorine gas dosing systems for chlorinating water via the indirect process work with negative pressure which is generated by an injector and reaches up to the chlorine tank valve.

If a dosing line that connects components of a chlorine system is damaged, no chlorine gas will escape. Ambient air will simply be sucked into the vacuum system.

Should the chlorine gas inlet valve of the vacuum regulator spring a leak, this just leads to slight overpressure in the dosing line system. This slight overpressure causes the pressure relief blow-off valve fitted in all vacuum regulators to open. The chlorine gas then flows in the blow-off line to the adsorption filter and the sensor of the gas detector.

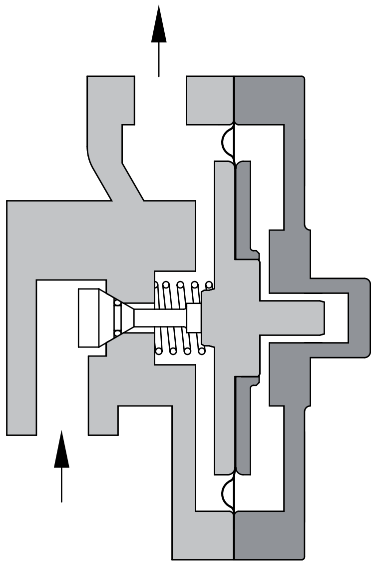
Should unfavourable circumstances cause an inlet valve and a dosing line or dosing device to spring a leak simultaneously, chlorine gas would escape uncontrollably, as the blow-off valve could not resond or at least not respond in full due to a lack of pressure.

To ensure that chlorine gas can under no circumstances escape outside the „chlorine gas storage room“, DIN 19606 requires that a pressure relief valve for preventing pressure in the vacuum line, i.e. a pressure relief safety stop valve, be fitted to the dosing line that continues into the next room (to the dosing regulator and further to the injector) before it exits the room.

6.2 Function

NOTE ***The safety stop valve is not a pressure relief safety stop valve and cannot be used to replace a pressure relief safety stop valve!***

After being started up, the injector generates a vacuum and drives the chlorine gas dosing system. This vacuum is then propogated through the diaphragm non-return, the vacuum dosing line and the dosing regulator right up to the safety stop valve. In the safety stop valve the vacuum causes the working diaphragm to move, while the mechanically coupled sealing cone releases the sectional area of flow.



The vacuum which now spreads ultimately opens the inlet valve via the diaphragm system in the vacuum regulator, and chlorine gas can then flow.

NOTE ***When in operation, the safety stop valve causes just a small additional loss of pressure due to the intentionally large working diaphragm. This drop in pressure is typically covered by the vacuum reserves already taken into account, and the safety stop valve then behaves like a length of dosing line.***

If the injector is switched off or if the vacuum fails for any other reason, spring pressure causes the safety stop valve to close too early against the weakening vacuum. The sectional area of flow is fully closed off and the flow of chlorine gas immediately interrupted.

Even with higher chlorine gas pressure on the vacuum regulator side of the stop valve, the valve is no longer able to open. The downstream system components remain depressurized.

NOTE ***The safety stop valve also prevents water from penetrating into the vacuum regulator if the diaphragm is broken in the diaphragm non-return of the injector or the sealing mechanism has a leak.***

6.3 Possible faults

NOTE ***Every time the safety stop valve is dismantled and then fitted again or a vacuum dosing line is replaced, a tightness check of the vacuum side must be performed!***

Error	Cause	Handling
Safety stop valve has a leak. Water can penetrate into the dosing line to the vacuum regulator.	Compression spring in the safety stop valve is broken	Remove the safety stop valve, dismantle it and remove the old compression spring. Clean the parts, fit a new compression spring and then refit the safety stop valve.
	Diaphragms are torn	Remove and dismantle the safety stop valve. Clean all parts. Unscrew the diaphragm disc and replace the diaphragm. Refit the diaphragm disc and stop valve.
	Heavy soiling at the valve Seat	Remove and dismantle the safety stop valve. Examine the valve seat extremely carefully and, if necessary, replace it (together with the O-ring on the valve taper seat). Refit the safety stop valve.
Desired dosing output not reached	Pressure losses in the dosing lines so severe that the suction power reserves of the injector are insufficient to cope.	Increase the cross section of the dosing lines. To do this, replace the input and output connections on the safety stop valve with larger ones. Alternatively, if possible shorten the dosing lines.
	Foreign bodies in the safety stop valve prevent the valve from opening.	Dismantle the safety stop valve and remove any foreign bodies. Refit the safety stop valve.
	Input and output sides of the safety stop valve have been switched. The valve cannot open.	Switch the input and output dosing lines.
	Injection is not working	Open the water isolating valve. Switch on the motive water (driving water) for the injector
	Adjustment spindle in the dosing regulator is closed.	Adjust the dosing regulator until the desired dosing is achieved.

7 Maintenance

Intervals for cleaning and maintenance

A visual check every six months is recommended.

The device should be dismantled, checked for soiling deposits and damage to its components, cleaned with warm water, dried and then refitted in an absolutely dry condition at least once a year by authorised staff.

We recommend performing maintenance every 2 years, during which the parts included in the maintenance kit should be replaced.

Should any components that are not included in the maintenance kit be defective, the safety stop valve should be returned to GRUNDFOS ALLDOS for repair.

WARNING ***Maintenance must always be performed by authorised specialist staff!***

NOTE ***Every time the safety stop valve is dismantled and then fitted again or a vacuum dosing line is replaced, a tightness check of the vacuum side must be performed!***

WARNING ***While the tightness check is being performed, all staff involved must wear protective clothing and gas masks!***

8 Spare parts

Maintenance kits

Order no.	**Use**
96729679 (553-943)	for chlorine gas
96729680 (553-944)	for ammonia gas and sulphur dioxide gas

8.1 Spare parts drawings

The parts labeled with item numbers are included in the spare parts sets. The maintenance kits each consist of a pressure spring and O-rings for the device and connections. FKM for chlorine gas and EPDM for ammonia gas and sulphur dioxide gas.